

Fungi and Viruses I

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Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Session Objectives

1. Compare and contrast fungal cells with animal and microbial cells
2. Describe the major types and characteristics of fungi
3. Outline the difference between yeast, mold and mushrooms
4. Define and explain 'fungal dimorphism'
5. Recognize the relationship between the environment and acquisition of fungal infections
6. State the different types of antifungal agents and their mechanism of action

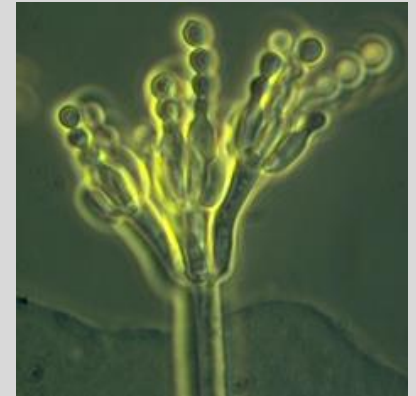
What are Fungi?

- Fungi are mostly 'free-living' eukaryotic cells
- Evolved alongside animal cells
- Non-motile, cell wall, non-photosynthetic
- Mostly very beneficial (saprophytes)

FEATURE	BACTERIA	FUNGI	VIRUSES
CELLS	YES	YES	NO
SIZE	1-5 MICRON	3-10 MICRON	0.02-0.2 MICRON
Genetic Material	DNA	DNA	DNA or RNA
Classification	PROKARYOTIC	EUKARYOTIC	N/A
RIBOSOMES	70S	80S	N/A
Membrane-bound organelles	NONE	YES	N/A
OUTER LAYER	PEPTIDOGYLCAN	CHITIN/GLUCAN	CAPSID/ENVELOPE
MOTILITY	SOME HAVE	NO	NO
REPLICATION	BINARY FISSION	BUDDING/MITOSIS	OTHER
Cell Membrane	PL/Cholesterol	PL/Ergosterol	N/A

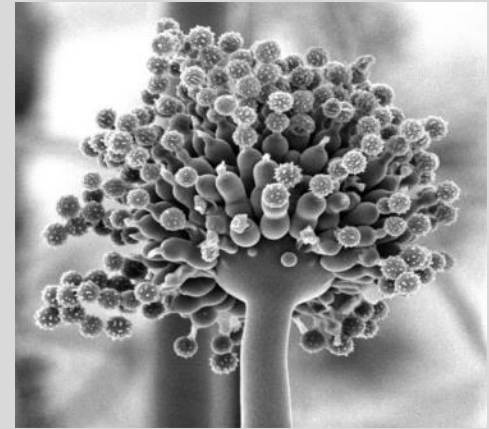
Fungi Characteristics

- Ubiquitous
- Unicellular and multicellular (***Dimorphic***)
- Mostly plant pathogens (humans are accidental hosts)
- Bread, beer & cheese
- Antibiotics (e.g., penicillin)
- Immunosuppressive drugs (e.g., cyclosporine)
- Aerobic/facultative anaerobes



Fungi emergence

- ↑ 1970s due to ↑ in IS individuals:
 - ✓ Cancer
 - ✓ AIDS
 - ✓ Transplant patients
 - ✓ Corticosteroid treatment
 - ✓ Catheters
 - ✓ Surgery
- Mycoses
- Prefer lower temp (20°C)
- Superficial mostly
- Systemic (↑ mortality)
- Treatment (challenging)



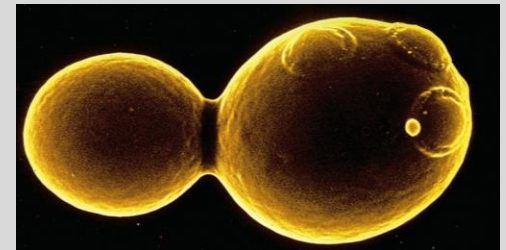
Aspergillus

Fungi: Basic Facts

- Pathogenic fungi are unicellular
- Must grow at 37°C for virulence
- Can be divided into 2 groups:
 1. Macroscopic (mushrooms)
 2. Microscopic
 - a) Yeasts
 - i. Spherical/elliptical shape
 - ii. Budding
 - b) Mold (hyphae)
 - i. Tubular branching filaments of fungal cells

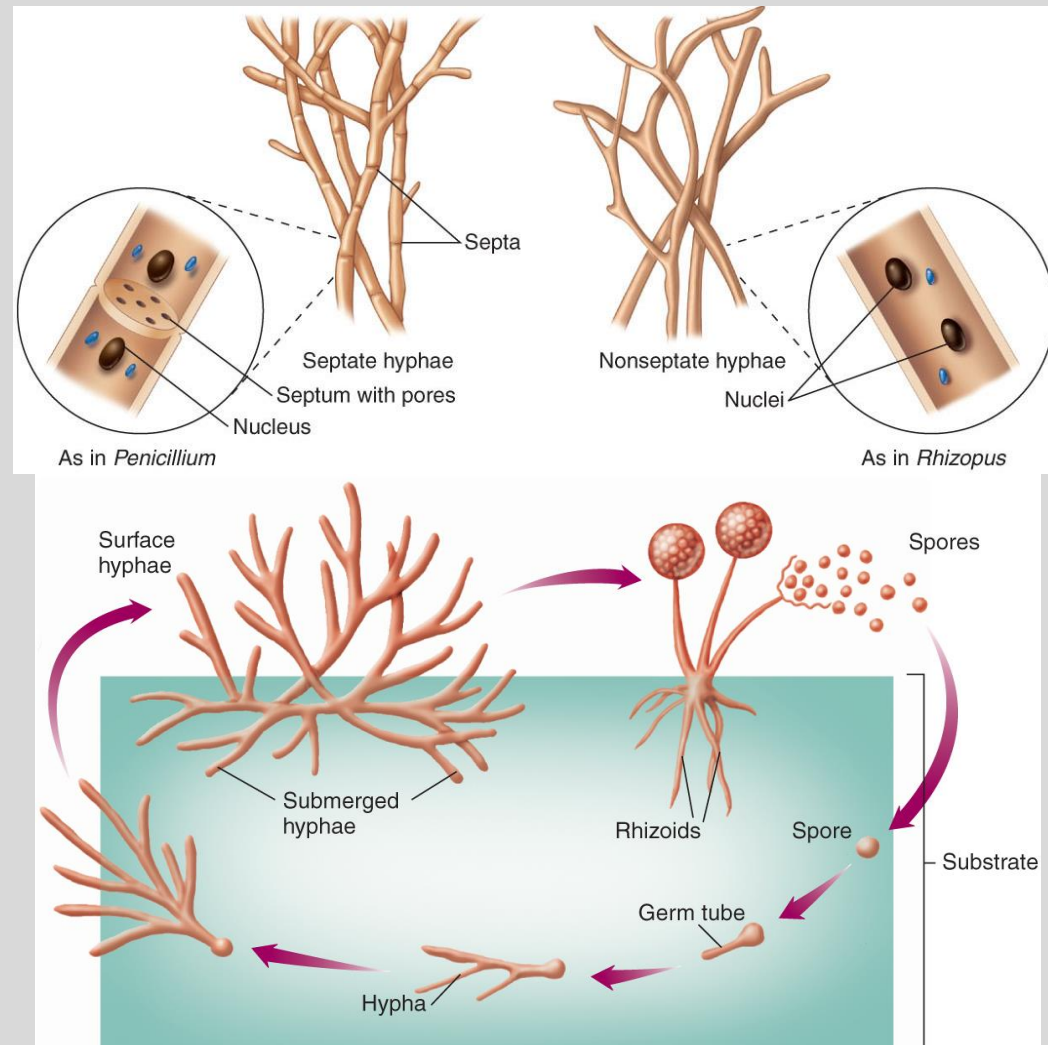


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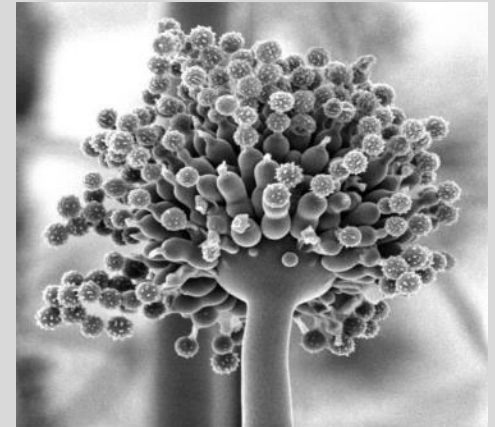
Hyphae

- Septate hyphae/
non-septate hyphae
- Mass of hyphae =
mycelium
- Vegetative hyphae -
anchor colony &
absorb nutrients
- Aerial hyphae -
project (spores)



Fungal Spores

- Derived from sexual or asexual reproduction
- Conidia (asexual spores) produce by pathogenic fungi
- Dormant, resistant, readily dispersed & germinate
- Ontogeny & morphology



Aspergillus

Acquisition of fungal infections

- Most from the environment
- Many do not penetrate intact skin
- Some are endogenous (*Candida*)
- Not contagious or transmissible
 - ✓ Exp.: Dermatophytes
- Type I Hypersensitivity
- Toxicity
- Epidemics reflect special conditions
 - ✓ Impaired host
 - ✓ Disturbance of ecosystems

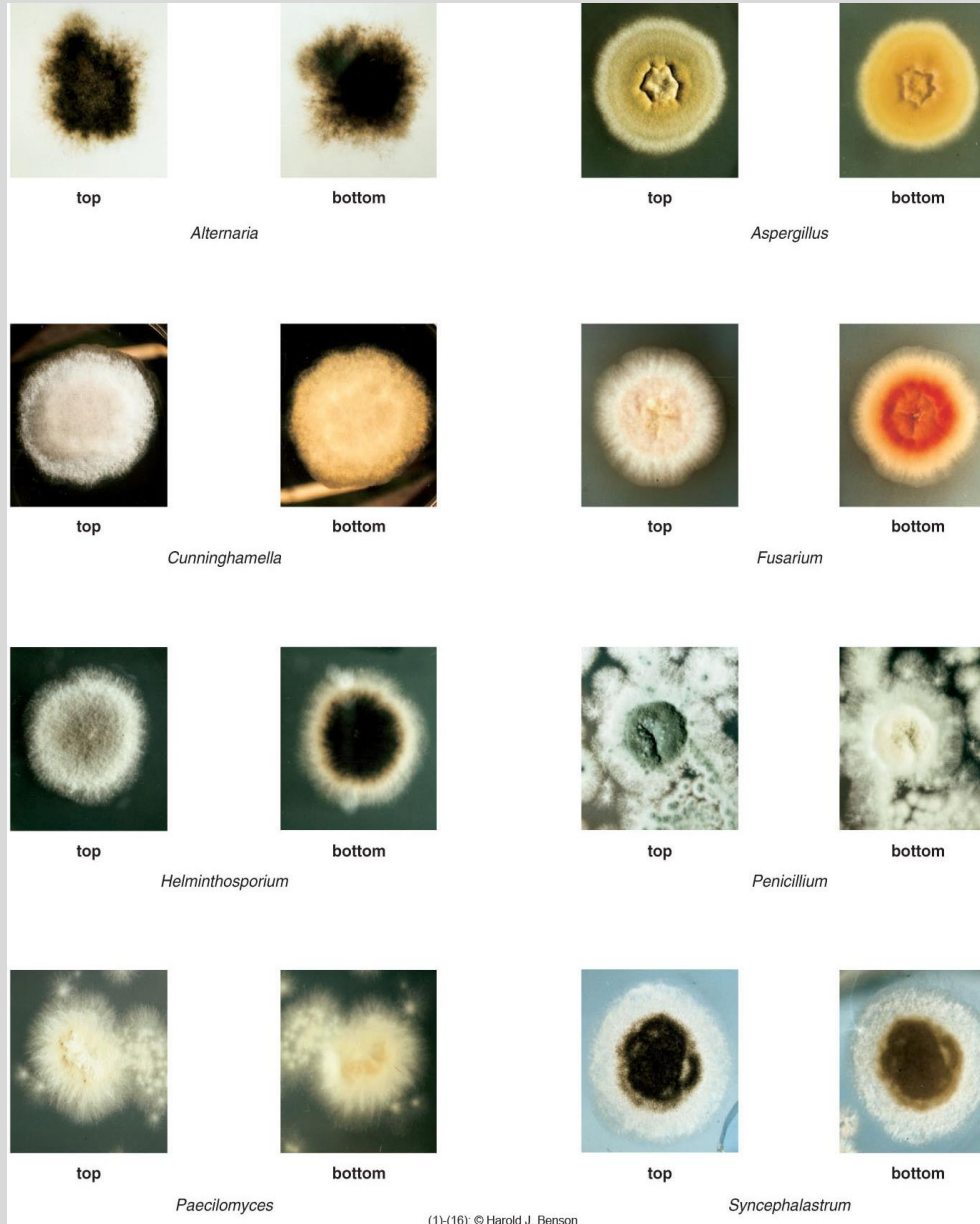
Diagnosis (Laboratory)

- Information combined with the site of infection
- Culture
 - ✓ Sabouraud's; Potato dextrose
 - ✓ Dimorphic (30 & 37°C)
- Microscopic examination (KOH; H&E)
- DNA-based detection (PCR; FISH)
- Serology

TABLE 45-2 Key Fungal Structures Observed in Microscopic Examinations of Clinical Specimens

Predominant Morphology	Mycoses
Yeasts—single or multiple buds	Blastomycosis, Histoplasmosis, Paracoccidioidomycosis, Penicilliosis, Sporotrichosis
Yeasts with capsules	Cryptococcosis
Hyphae—septate	Hyalohyphomycosis—species of <i>Aspergillus</i> , <i>Fusarium</i> , <i>Geotrichum</i> , <i>Trichosporon</i> , et al.)
Hyphae—septate in skin or nail specimens	Dermatophytosis
Hyphae—nonseptate	Mucormycosis—species of <i>Rhizopus</i> , <i>Lichtheimia</i> , <i>Cunninghamella</i> , et al.
Hyphae—septate; brownish cell walls	Phaeohyphomycosis—species of <i>Bipolaris</i> , <i>Cladosporium</i> , <i>Curvularia</i> , <i>Exserohilum</i> , et al.
Yeasts and Pseudohyphae	Candidiasis—species of <i>Candida</i>
Yeasts and Hyphae in skin scrapings	Pityriasis versicolor
Spherules	Coccidioidomycosis
Sclerotic cells—brownish cell walls	Chromoblastomycosis
Sulfur granules	Mycetoma
Arthroconidia in hair	Dermatophytosis
Conidia in pulmonary cavity	Hyalohyphomycosis—species of <i>Aspergillus</i> , <i>Fusarium</i> , et al.
Cysts (asci) in pulmonary specimens	<i>Pneumocystis</i>

Colonial Morphology



Fungal epidemics

- California earthquakes result in increased Coccidioidomycosis
- Construction near hospitals increase *Aspergillus fumigatus* infections
- Tree cutting associated with outbreaks of Histoplasmosis
- Epidemiology reflects the presence of fungus in the environment & the immunological state of the host
 - Histoplasmosis in the Ohio River Valley
 - Coccidioidomycosis in the Southwest

Mycoses

- Superficial (non-living components)
- Cutaneous mycoses
- Subcutaneous mycoses
- Systemic mycoses

- Opportunistic mycoses

Superficial & Cutaneous mycoses

- Most common of all communicable diseases
- Most caused by *Malassezia*, dermatophytes, or *Candida* (normal flora)
- Dermatophytes are inhibited by body temperature; rarely become invasive

✓Types:

- *Microsporum*
- *Trichophyton*
- *Epidermophyton*

✓Persistent & troublesome

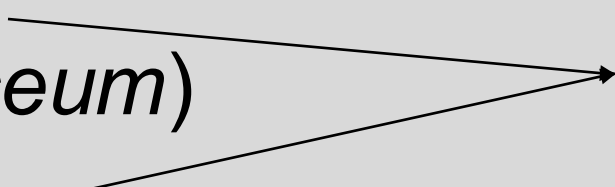
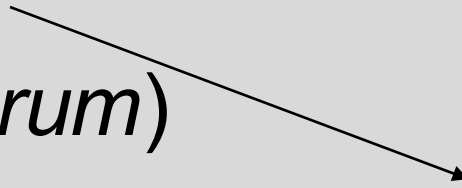


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Dermatophytes

1. Geophilic (soil;
Microsporum gypseum)
 1. Acute
 2. resolves quickly
 3. Rarely recur
2. Zoophilic (animals;
Microsporum canis)
3. Anthropophilic (humans;
most common;
Trichophyton rubrum)
 1. Chronic
 2. Prolonged treatment
 3. Recurs frequently

Subcutaneous Mycoses

- Reside in soil or vegetation
- Cut → granulomatous nodules or ulcers → expand slowly
- Confined to subcutaneous layer
- E.g.: *Sporothrix schenckii*

Sporotrichosis

Traumatic inoculation of *Sporothrix schenckii*
Found in **plants** and in **soil** (rose thorns)



Suppurating **subcutaneous nodules**
Progresses proximally along **lymph channels**
Rx: **Itraconazole**

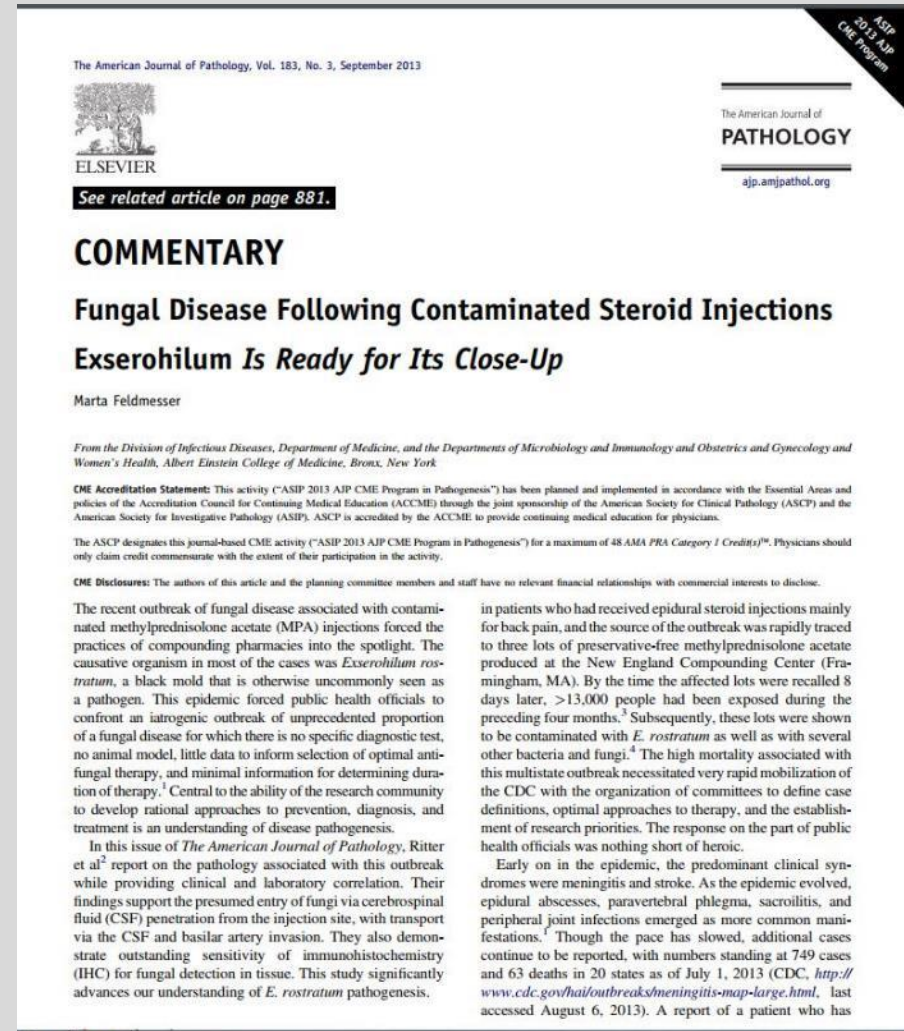
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Systemic Mycoses

- Soil saprophytes (mostly inhaled)
- Opportunistic:
 1. Candidiasis (most common)
 2. Cryptococcosis
 3. Aspergillosis
 4. Mucomycosis
 5. *Pneumocystis* pneumonia
- Endemic:
 1. Coccidioidiomycosis
 2. Histoplasmosis
 3. Blastomycosis

The outbreak of *Exserohilum rostratum*

- September 2012
- Contaminated methylprednisolone (epidurals)
- 13,000 patients exposed; 750 developed illness; 63 died

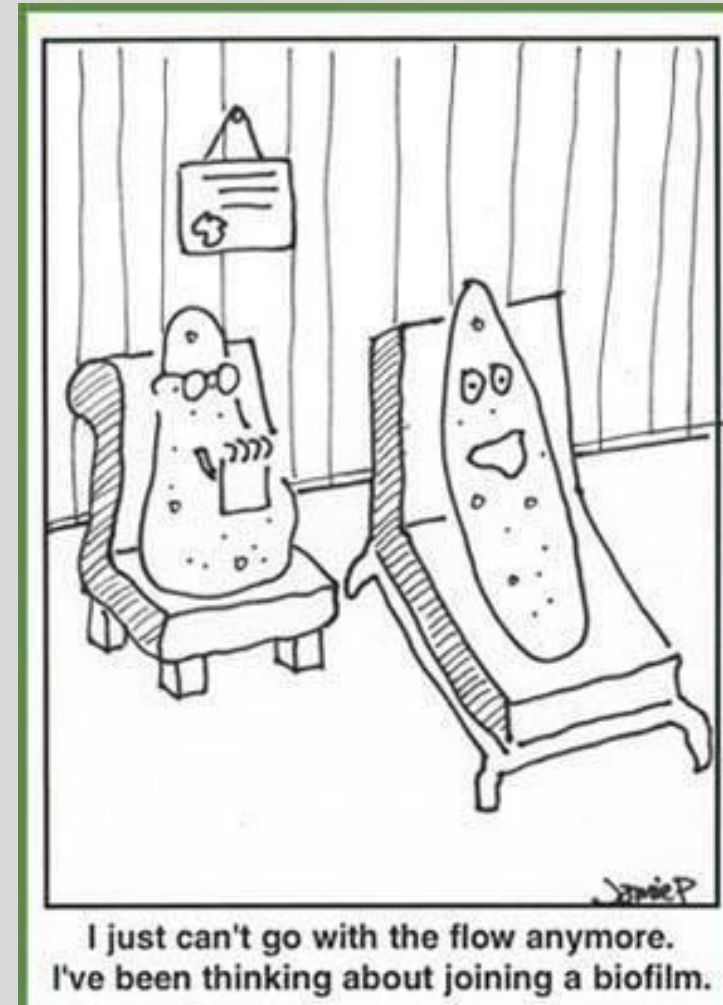


Medical pathogenic fungi

Degree of Tissue Involvement and Area Affected	Name of Infection	Name of Causative Fungus
Superficial (not deeply invasive)		
Outer epidermis	Tinea versicolor	<i>Malassezia furfur</i>
Epidermis, hair, and dermis can be attacked.	Dermatophytosis, also called tinea or ringworm of the scalp, body, feet (athlete's foot), toenails	<i>Microsporum</i> , <i>Trichophyton</i> , and <i>Epidermophyton</i>
Mucous membranes, skin, nails	Candidiasis, or yeast infection	<i>Candida albicans</i>
Systemic (deep; organism enters lungs; can invade other organs)		
Lung	Coccidioidomycosis (San Joaquin Valley fever)	<i>Coccidioides immitis</i>
	North American blastomycosis (Chicago disease)	<i>Blastomyces dermatitidis</i>
	Histoplasmosis (Ohio Valley fever)	<i>Histoplasma capsulatum</i>
	Cryptococcosis (torulosis)	<i>Cryptococcus neoformans</i>
Lung, skin	Paracoccidioidomycosis (South American blastomycosis)	<i>Paracoccidioides brasiliensis</i>

Fungal virulence factors

- Genetic switching of metabolic processes
- Production of adhesions
- Enzymes: Proteases, catalase, nuclease, phospholipase, etc.
- Phagocytosis: glucan, melanin, capsules
- Biofilm formation



Fungal allergies/toxins

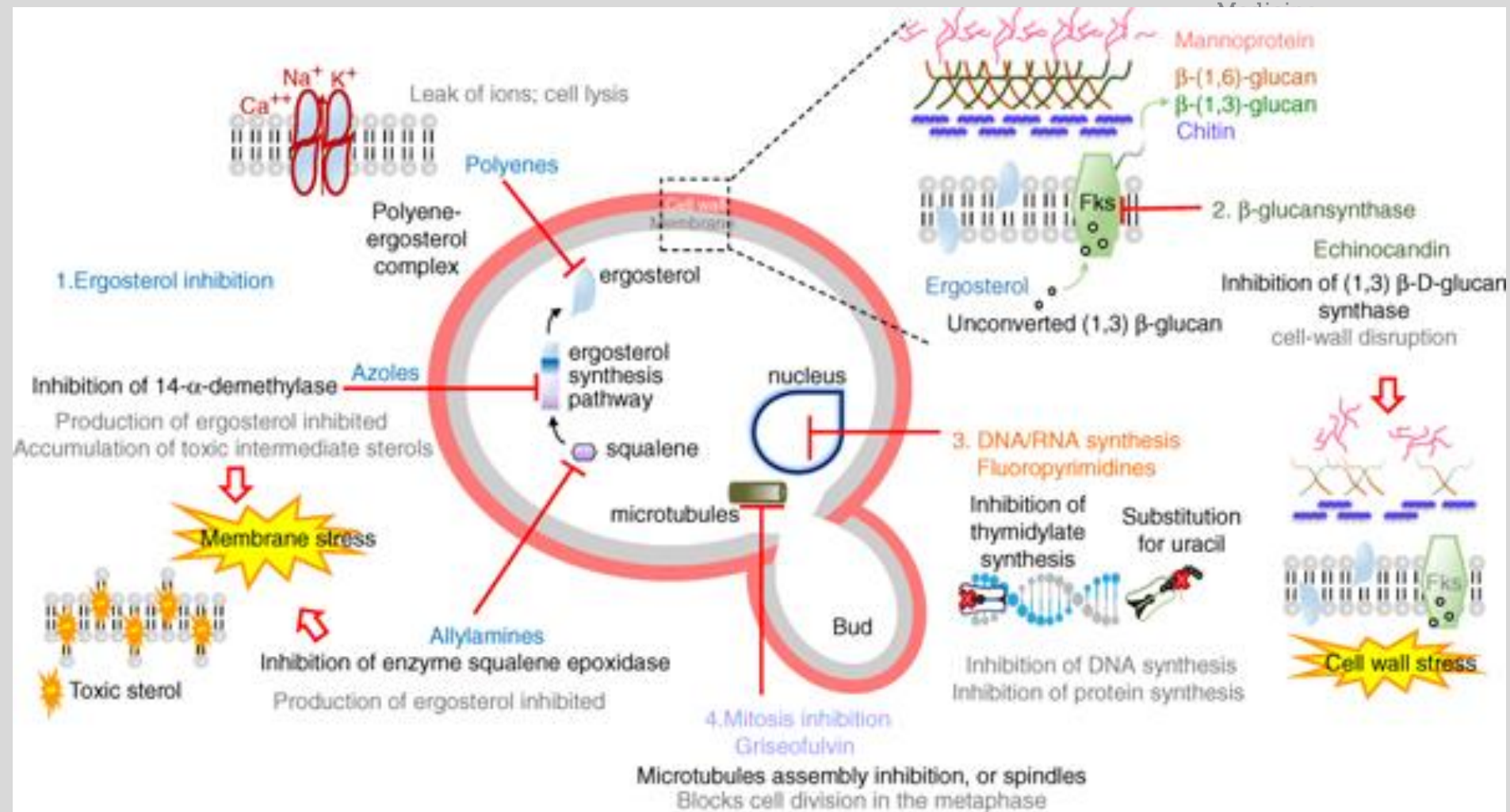
- Type I hypersensitivity
- Caused by non-infectious mold
- Outdoor or indoor (sick building syndrome)
- Mycotoxins (mushrooms; aflatoxin, α -amanitin)

Antifungal agents

- Polyenes: amphotericin B (broad spectrum) & nystatin (topical); bind ergosterol
- Flucytosine: pyrimidine analogue
- Azoles: ketoconazole, itraconazole, fluconazole, miconazole (topical) etc.; inhibit ergosterol synthesis
- Griseofulvin: microtubule disruption; dermatophytes
- Terbinafine: inhibit ergosterol synthesis
- Echinocandins: caspofungin/micafungin; inhibit glucan synthesis
- mAbs to fungal Ags



Antifungal targets



Summary Slide

- Fungal cells with human counterparts and other microbes
- Described the major types and characteristics of fungi
- Outlined the difference between yeast and mold.
- Described 'fungal dimorphism'
- The relationship between the environment and acquisition of fungal infections
- Antifungal agents

References

1. Murray, P.R., Rosenthal, K. S., and Pfaller, M.A. 2021. **Medical Microbiology**. Elsevier Inc.
2. K. C. Carroll et. al. 2016. **Jawetz, Melnick, & Adelberg's Medical Microbiology**, 27th ed. McGraw-Hill Education. ISBN 9780-0-71-82498-9.
3. Tao, LE, V. Bhushan and M. Sochat. 2024. **First Aid for the USMLE Step 1**; McGraw Hill Publication.

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>